

Multiple quasi-phase-matching in a nonperiodic domain-inverted optical superlattice

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We theoretically propose a nonperiodic optical superlattice (NOS), in which multiple nonlinear optical parametric processes can be realized simultaneously. The optimal design of the NOS structure is achieved by the combination of stimulated annealing (SA) and genetic algorithm methods. The multiple quasi-phase-matched (QPM) second harmonic generations (SHG's) in NOS's are theoretically investigated. The numerical simulations show that NOS's can be used as a more effective and useful nonlinear optical crystal for multiple nonlinear optical parametric generation.

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With the development of the electric poling technique at room temperature, it is possible to build various optical superlattices in ferroelectric crystals such as LiTaO_3 , LiNbO_3 , etc., with periodic and quasiperiodic domain-inverted structures. Nonlinear optical interactions between wave vectors of interactive waves and reciprocal vectors of the optical superlattice have led to laser frequency generations with quasi-phase-matching (QPM) schemes. The QPM, which has been around for a long time [1], uses periodic modulation of the nonlinear coefficient of crystal to compensate for the mismatch among the wave vectors of the interacting light beams [2–6], therefore, it substantially extends the class of materials available to various nonlinear optical interactions. This first kind of superlattice called the periodic optical superlattice (POS) provides only one reciprocal vector to compensate for the wave-vector mismatch in a given nonlinear optical process, and has a narrow acceptance bandwidth for pump wavelength and operating temperature. In order to achieve multiple QPM schemes, quasiperiodic optical superlattice (QOS) [7,8], such as the Fibonacci structure, was suggested. Quasi-phase-matched third-harmonic generation (THG) and multiple second harmonic generation (SHG) have been experimentally demonstrated. Then came the theoretical predication of the aperiodic optical superlattice (AOS) [9,10], which can supply many more reciprocal vectors for multiple QPM's than POS and QOS. Experimental observations of multiple QPM processes have been reported recently [11]. In addition, Chou *et al.* [12] reported difference frequency generation-based wavelength converters with multiple phase matching in LiNbO_3 waveguides by employing a phase-reversal sequence superimposed upon a uniform QPM grating. The above approaches expand the amount of the requisite superlattice reciprocal vectors for QPM's greatly and make multiple QPM nonlinear optical processes available.

In the AOS theory, in order to achieve a predesigned multiple QPM process, samples are divided into blocks with congruent length. The spontaneous polarization direction of each block is determined by the simulated annealing (SA) method to optimally construct the structure. Although the periodicity of domain inversion disappears in AOS optical superlattices, each block must be of one or multiple fixed

lengths. So, it is naturally expected that the arbitrary domain length of a block in optical superlattice will introduce even more and flexible reciprocal vectors for QPM nonlinear optical processes than the AOS. Following the work mentioned above, we propose a kind of optical superlattice: nonperiodic optical superlattice (NOS) in which the limitation to block length disappears. In this paper, we discuss the advantage of NOS in nonlinear optical frequency conversion. We believe that this kind of superlattice may also have potential advantages for the improvement of the properties of quadratic solitons [13–17].

Here, we use a lithium niobate crystal as an example to demonstrate NOS. A lithium niobate sample with NOS structure is shown in Fig. 1. It is divided into many blocks with the random length and domain orientation along the propagation direction. In order to use the largest nonlinear coefficient d_{33} , the interfaces of each domain are parallel to the YZ plane and the propagation and polarization directions of incident light are along the X and Z axes, respectively. A laser beam is launched into the crystal from the left side and SHG is generated in the sample through a QPM nonlinear optical process. In a so-called small signal approximation, the conversion efficiency η from fundamental wave to second harmonic wave reads

$$\eta_{\text{SHG}} = \frac{8\pi^2 |d_{33}|^2 I_\omega L^2}{c \epsilon_0 \lambda^2 n_{2\omega} n_\omega^2} \left| \frac{1}{L} \int_0^L dx e^{i(k_{2\omega} - 2k_\omega)x} \tilde{d}(x) \right|^2, \quad (1)$$

where k_ω ($k_{2\omega}$) is the wave number of the fundamental (second harmonic) wave, c is the speed of light in vacuum, n_ω

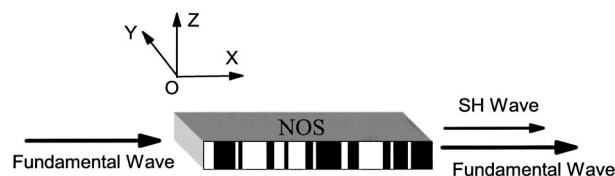


FIG. 1. Schematic diagram of the lithium niobate sample with NOS structure. The black (white) strips represent the positive (negative) domains. Fundamental wave and SH wave propagation is along the X direction while their polarization is along the Z direction.

($n_{2\omega}$) represents the refractive index of the fundamental (second harmonic) wave, λ is the fundamental wave in vacuum, ε_0 is the permittivity of vacuum, L is the total length of the sample, and $\tilde{d}(x)$ represents the orientation of each block taking binary values of 1 or -1 .

One of the goals in studying optical superlattice is to find proper domain structures in which multiple predefined nonlinear optical processes can be implemented. In AOS [9,10], samples are divided into blocks with congruent length, and the spontaneous polarization direction of each block can be determined by a simulated annealing (SA) method to optimally construct the structure. But in NOS, the SA algorithm is no longer effective because the length of each block is not fixed.

To simplify the problem of designing a NOS structure in lithium niobate, we just consider the integral part in the Eq. (1) that represents the effective nonlinear coefficient of optical superlattices:

$$d_{\text{eff}} = \frac{1}{L} \left| \int_0^L dx e^{i\alpha x} \tilde{d}(x) \right|, \quad (2)$$

where d_{eff} is an effective nonlinear coefficient, representing the SHG conversion efficiency, and $\alpha = k_{2\omega} - 2k_{\omega}$. The sample is divided in blocks with the number of N and the position of each block is located between x_q and x_{q+1} , for $q = 0, 1, 2, 3, \dots, N-1$. We evaluate this integral (2) as

$$d_{\text{eff}} = \frac{1}{L} \sum_{q=0}^{N-1} \tilde{d}(x_q) \int_{x_q}^{x_{q+1}} dx e^{i\alpha x}. \quad (3)$$

It is clearly seen that the contributions to d_{eff} are from two factors: the length and domain orientation of each block. In AOS theory, we can easily optimize the domain structure for SHG with the SA method because the length of each block is congruent. But in the case of NOS, we need to determine not only the domain orientation of each block but also the length of each block. Here we use a combination of two algorithm methods, SA and the genetic algorithm [18,19], to solve the structure optimization problem in NOS. We call it the SG algorithm method.

Figure 2 shows a flow chart of the SG method for constructing NOS in lithium niobate. The procedure of this algorithm is described in detail as follows.

(1) Define some conditions and parameters used in the SA method, such as initial temperature, dropping rate of temperature, and target functions according to a predefined nonlinear optical parametric process. (The temperatures defined above are the parameters used in the SA algorithm, not the working temperature of the NOS crystal.)

(2) Define the group's size, i.e., the number of individuals in the group. Each individual represents one kind of NOS structure.

(3) The number of each individual's domain blocks is N and each block's position x_q in all the individuals is set by a random value generated by a computer.

(4) Based on the target function set in procedure 1, we achieve the spontaneous polarization direction of each block

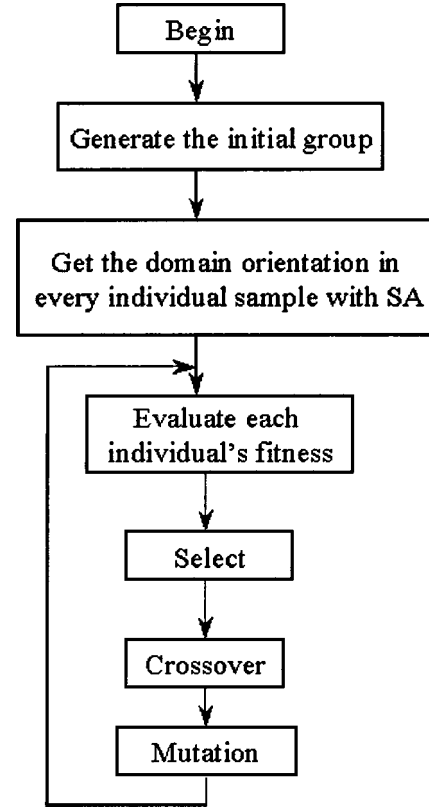


FIG. 2. Flowchart of the SG algorithm for the design of NOS.

in every individual by SA method. Till now the initial NOS structure of each individual is obtained and the initialization of the group is accomplished.

(5) Define a new target function (the target function may be different than that in procedure 1). Evaluate each individual according to the new target function, then select the better individuals from the former group to construct a more favorable new group.

(6) Implement the crossover and mutation procedure defined in the genetic algorithm.

(7) Continue procedures 5 and 6 until the acceptable NOS structure for predefined nonlinear optical parametric process appears.

To demonstrate the effectiveness of the SG method in optimizing the NOS structure for a predefined nonlinear optics process, we consider a multiple QPM SHG process in NOS that can achieve multiple wavelength SHG's with identical effective nonlinear coefficient $d_{\text{eff}}(\lambda_{\alpha}) = d^{(0)}$, where $d^{(0)}$ is a constant. The four wavelengths λ_{α} 1.0600, 1.0820,

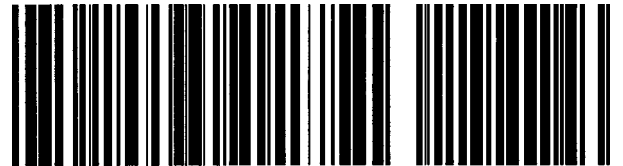


FIG. 3. A gray-scale diagram of the NOS in part. The black (white) strips represent the positive (negative) domains. Four QPM SHG process can be implemented with fundamental wavelengths of 1.0600, 1.0820, 1.283, and 1.3640 μm .

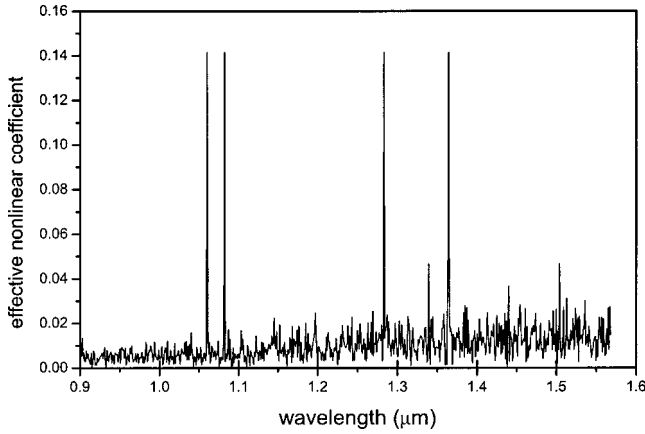


FIG. 4. Effective nonlinear coefficient of QPM SHG as a function of fundamental wavelength in NOS structure optimized by the GA algorithm with $N=3000$. Four peaks with almost identical effective nonlinear coefficients of about 0.14 at the designed wavelengths are found.

1.283, 1.3640 μm are chosen as the fundamental waves. The dispersions of the refractive index of material are evaluated according to the Sellmeier equation [20]. The objective function in the GA method is chosen as

$$E = \sum_{\alpha} |d^{(0)} - d_{\text{eff}}(\lambda_{\alpha})| + [\max\{d_{\text{eff}}(\lambda_{\alpha})\} - \min\{d_{\text{eff}}(\lambda_{\alpha})\}], \quad (4)$$

where $\max\{\dots\}$ and $\min\{\dots\}$ take their maximum and minimum values among all the quantities in $\{\dots\}$, respectively.

Figure 3 is a grayscale diagram of the constructed NOS in part from the calculated result. The black (white) strips represent the positive (negative) domains. The constructed structure of the NOS is no longer a periodic structure and the length of each block is totally irregular. In our simulation, the maximum and minimum block length is 1.2 and 678.2 μm and the minimum difference between two domains is 0.2 μm . Figure 4 displays the calculated effective nonlinear coefficient with the optimized nonperiodic structure as shown in Fig. 3 after scanning a wide range of wavelength. Four strong peaks with almost identical peak values of about 0.14 at the pre-designed wavelengths are achieved only at the expense of the reduced effective nonlinear coefficient lower than that of the perfectly periodic QPM grating. The effective nonlinear coefficient decrease in NOS is due to the trade off between the multiple QPM and conversion efficiency. For perfect QPM uniform grating, the ideal value of the effective nonlinear coefficient is 0.6366 ($2/\pi$). The sum of the effective nonlinear for the desired peaks in NOS is 0.56, about 88% of the ideal value of the effective nonlinear coefficient.

NOS constructed using the SG algorithm has obvious advantages in multiple QPM nonlinear optics processes. First, a more powerful algorithm SG method which is the combination of the SA and genetic algorithms is used in solving the optimization nonlinear optical problem in NOS. The SA method is not good at getting the best local results and the genetic algorithm can be easily used to guide the search pro-

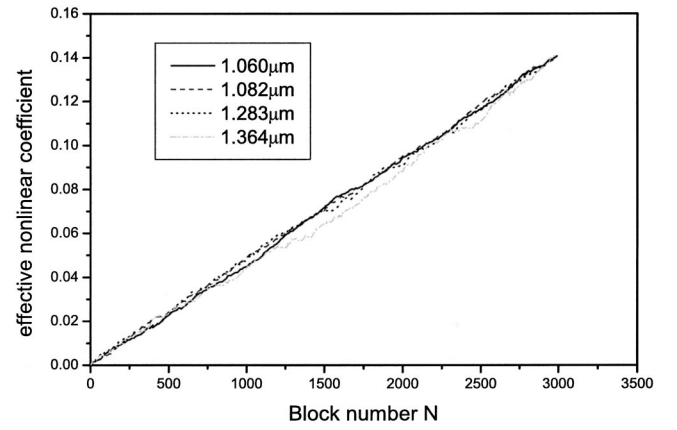


FIG. 5. Display of the SHG effective nonlinear coefficient of four fundamental wavelengths as they propagate in the NOS sample. The SHG efficiency of the four wavelengths is identical at the end but the evolution process of SHG for four wavelengths along the propagation in NOS is different.

cess towards the correct directions. So, the SG method that takes advantage of both the SA and genetic algorithms can construct more suitable and flexible superlattices for multiple QPM's. Second, because there are two different target functions in the SG, more complex searching demands and various nonlinear problems can be implemented. Additionally, in the AOS structure the length of the basic block should first be defined, and in the NOS this kind of restriction can be totally avoided so we can get more favorable and flexible reciprocal vectors.

An identical effective nonlinear coefficient for four fundamental waves in NOS as shown in Fig. 4 implies that SHG efficiency is the same when the four fundamental waves propagate through the whole length of the crystal. Because the responses to each block inside the NOS depend on the wavelength of light beams, we can suppose that the accumulative processes for SHG power along the crystal are different. Figure 5 describes the evolution of the effective nonlinear coefficient of four fundamental waves described by Eq. (3) as they propagate through series domain blocks in the NOS. From the figure, we can see that each curve moves to the same end point in different ways. This clearly shows that these wavelengths can all be globally phase matched by the interference effect of all constructed domains with the same effective nonlinear coefficient, but undergo different interference processes.

In summary, we have proposed an optical superlattice NOS to realize multiple second harmonic generations simultaneously. With the combination of SA and genetic algorithms, we obtain an ideal construction of NOS for pre-designed multiple QPM SHG processes. The calculated results show that a nonlinear crystal with NOS structure is more favorable and flexible in nonlinear optical harmonic generations.

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